Methods 8

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Apologies for the late submission! You can view my repo's commit history here if that helps to verify that this is my own work done independent from the solutions.

1.

a.

$$\text{Provided that :} \qquad \bar{Y}(A=0,B=0)=1, \qquad \bar{Y}(A=1,B=0)=5,$$

$$\bar{Y}(A=0,B=1)=4, \qquad \bar{Y}(A=1,B=1)=7$$

we can hold the following to be true.

$$\begin{aligned}\bar{Y}(A=0,B=0) &= \mathbb{E}(Y_i \mid A=0,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 0+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+0+0+0+\mathbb{E}(\epsilon_i)\\ &= \beta_0+0=1\\ &\therefore \beta_0=1\end{aligned}$$

$$\begin{aligned}\bar{Y}(A=1,B=0) &= \mathbb{E}(Y_i \mid A=1,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 1+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+\beta_1+0+0+\mathbb{E}(\epsilon_i)\\ &= 1+\beta_1+0=5\\ &\therefore \beta_1=4\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=0,B=1) &= \mathbb{E}(Y_i \mid A=0,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \beta_3 \cdot 0 + \epsilon_i) \\
&= \beta_0 + 0 + \beta_2 + 0 + \mathbb{E}(\epsilon_i) \\
&= 1 + \beta_2 + 0 = 4 \\
&\therefore \beta_2 = 3
\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=1,B=1) &= \mathbb{E}(Y_i \mid A=1,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 1 + \beta_3 \cdot 1 + \epsilon_i) \\
&= \beta_0 + \beta_1 + \beta_2 + \beta_3 + \mathbb{E}(\epsilon_i) \\
&= 1 + 4 + 3 + \beta_3 = 7 \\
&\therefore \beta_3 = -1
\end{aligned}$$

Final Answer: $\boxed{\beta_0 = 1, \beta_1 = 4, \beta_2 = 3, \beta_3 = -1}$